

Einführung in die Kern-, Teilchen und Astrophysik

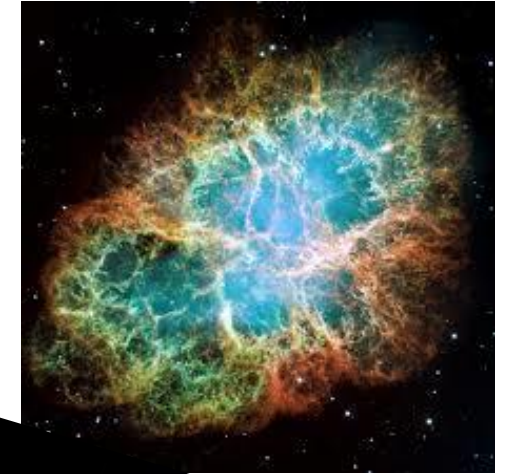
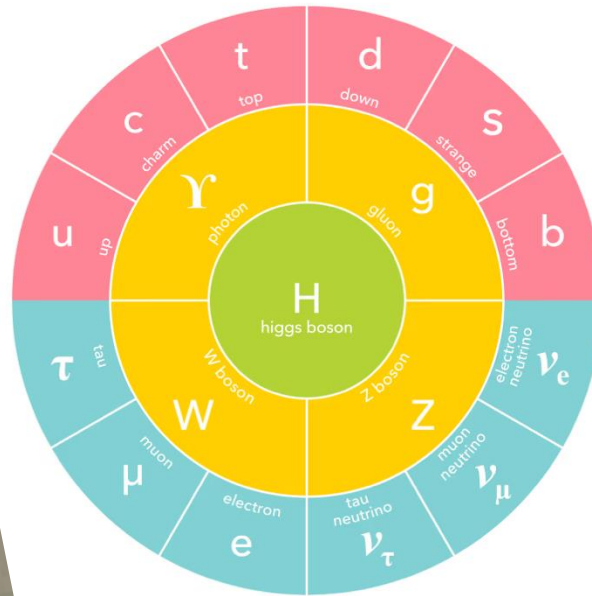
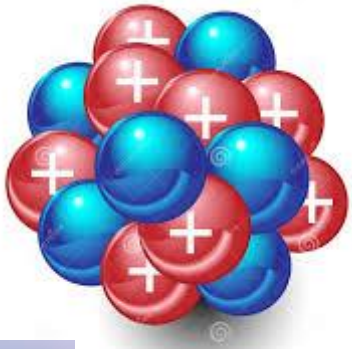
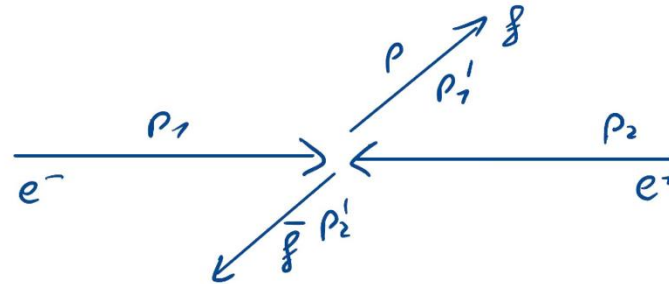
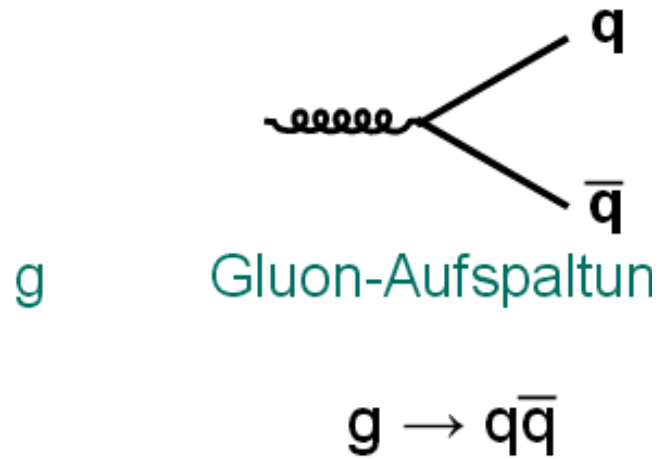


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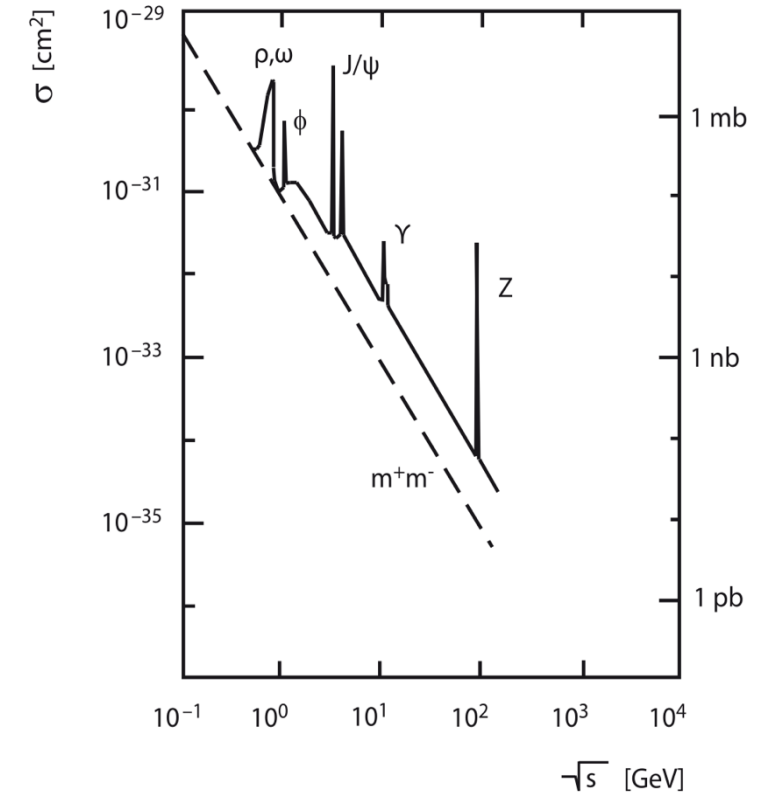
Lecture	Date	Theme	Topic	Sub-Topics
1	14/10/25	Introduction	Little History of particle physics	atoms, elementary particles
2	21/10/25	Particle Physics	Standard Model	Fermions, Bosons, Interactions,
3	28/10/25		Fundamental symmetries	Conservations, Feynman, Propagator,
4	04/11/25		Fundamental symmetries	Discrete symmetries; Cross section
5	11/11/25		Interaction with Matter	Bethe Bloch, gamma interaction
6	18/11/25		Particle detection	Gamma interactions, detector basics
7	25/11/25	Astrophysics	Detectors, accelerators	Detector types, accelerators, cyclotron Supernovae, cosmic rays, Supernovae, cosmic rays
8	02/12/25	Nuclear physics	Basics	Binding energies, Weizäcker
9	09/12/25		Nuclear reactions	Beta, alpha, fission
10	16/12/25		Nuclear reactors	Radioactive transitions, Radiocarbon method
11	13/01/26	Particle / Nuclear	Nuclear Structure	Gamma transitions, Mott, Form factors
12	20/01/26		Nucleon Structure	Deep inelastic scattering, Quarks and QCD
13	27/01/26	Particle Physics	Weak interactions	Quarks and QCD (VOL2), Z ₀ production, Parity violation
14	03/02/26	Astrophysics	Neutrinos	Neutrino basics, Neutrino oscillations



Zusammenfassung der letzten Vorlesung



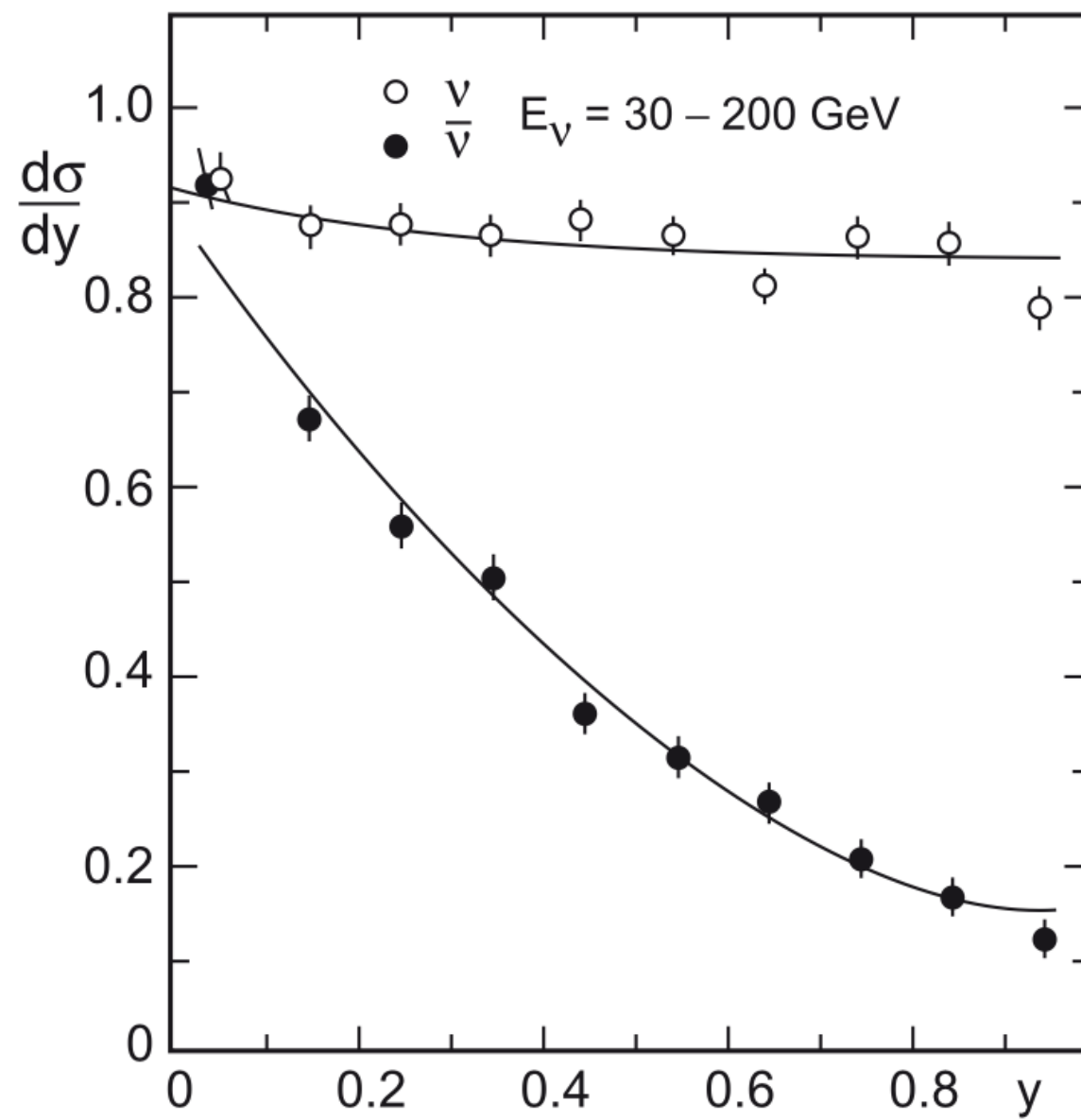
$$\begin{aligned} \underline{s} &= m_1^2 c^4 + m_2^2 c^4 + 2E_1 E_2 - 2\vec{p}_1 \vec{p}_2 c \\ &= 2E^2 + \underbrace{2p_1^2 c^2}_{\sim 2E^2} \approx \underline{4E^2} \end{aligned}$$



Farbladungen von Quarks
und Gluonen

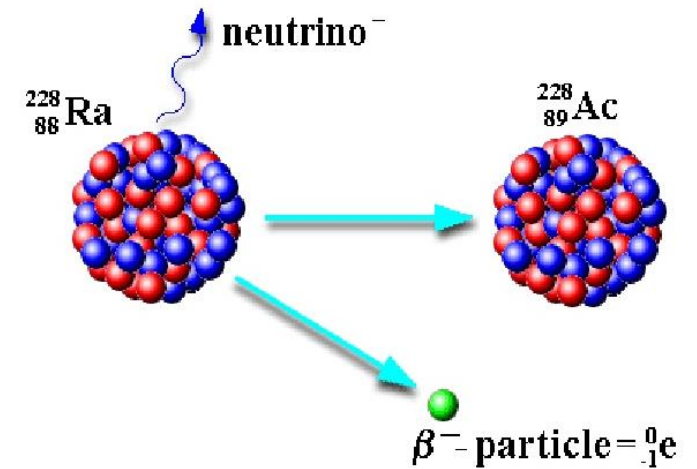
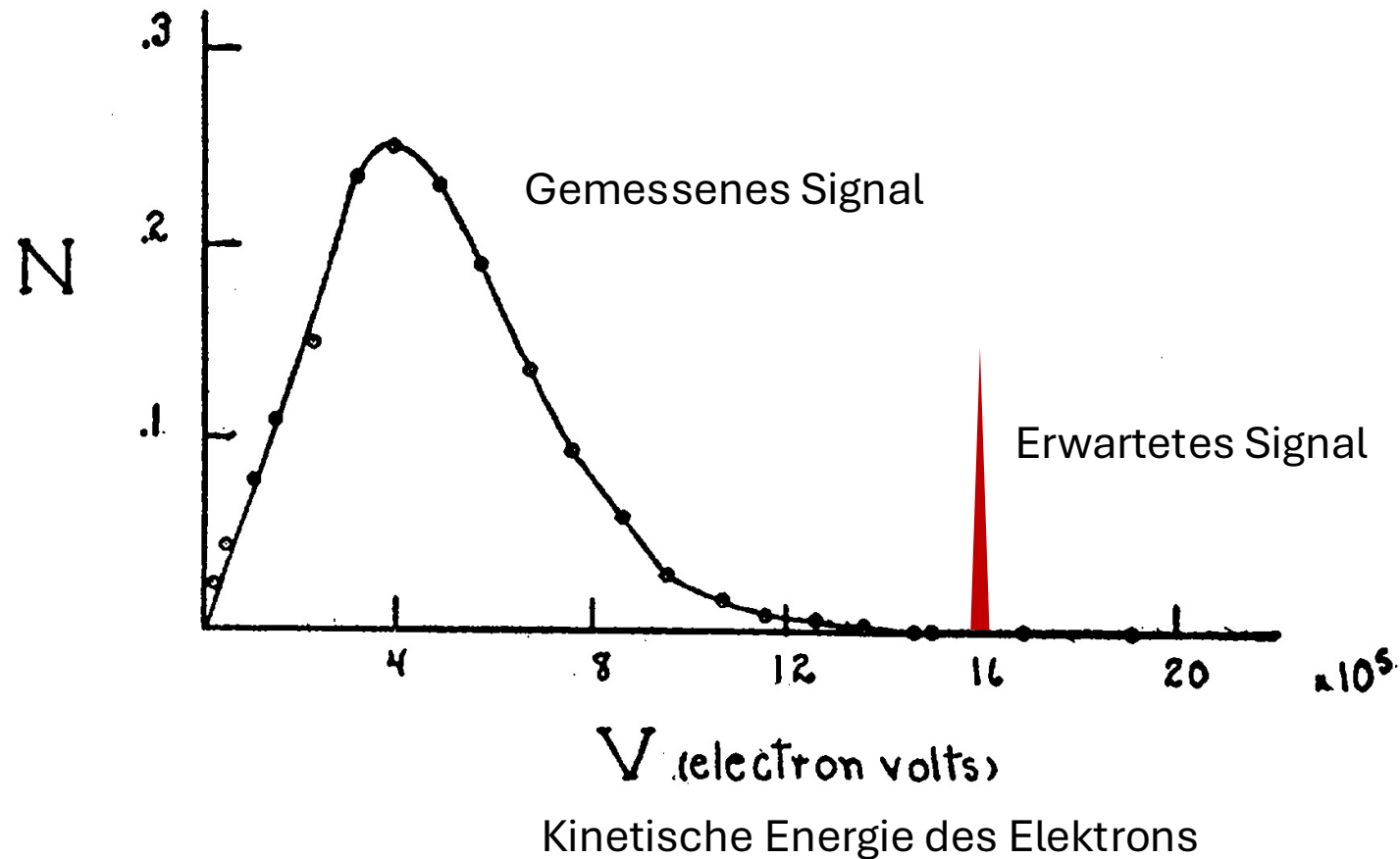
e⁺ e⁻ Kollisionen

Resonanzen



Poch Fig. 10.2

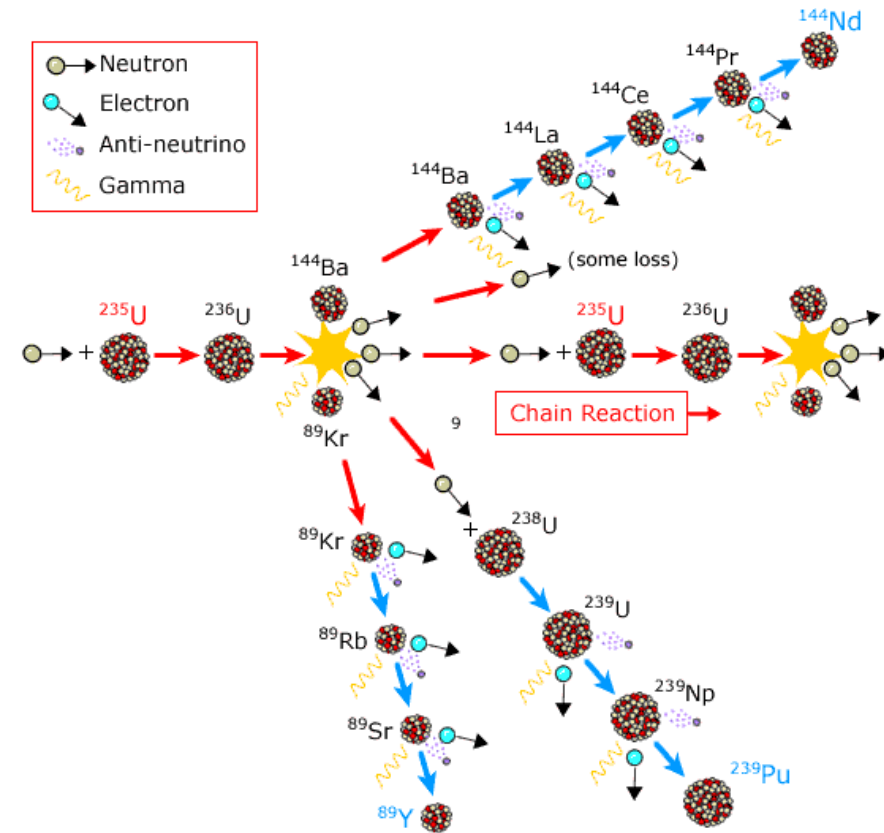
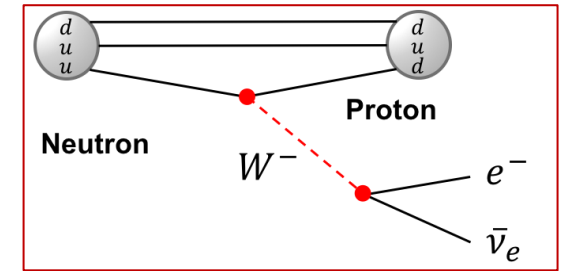
Warum Neutrinos? β -Zerfall von Radium



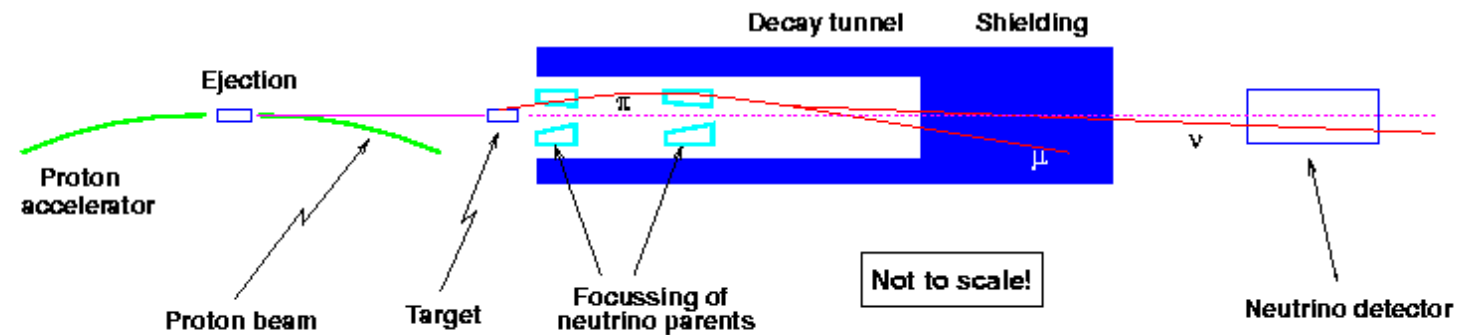
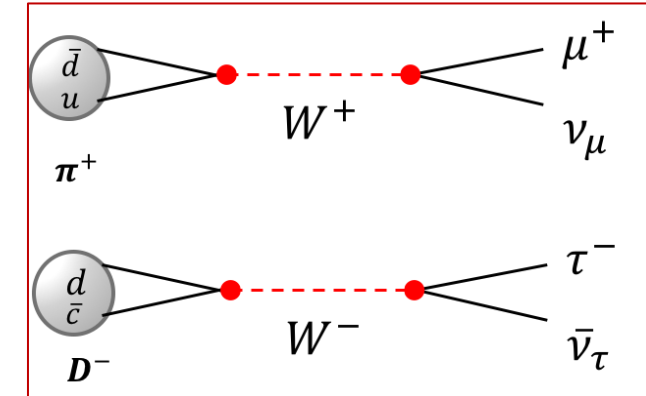
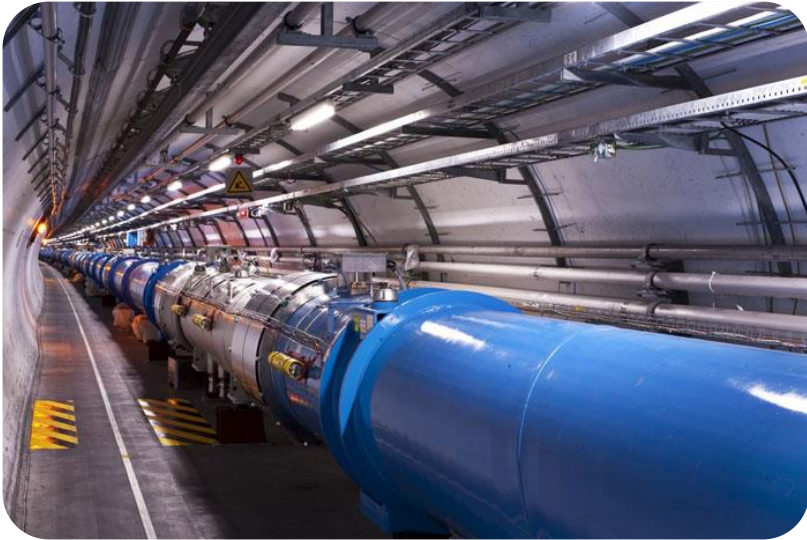
Neutrinos aus Kernreaktoren



- Kerspaltung von ^{235}U und ^{239}Pu
- Beta Zerfall Kaskaden von neutronenreichen Spaltproduktion
- Im Mittel 6 Neutrinos per Spaltung



Neutrinos aus Beschleunigern

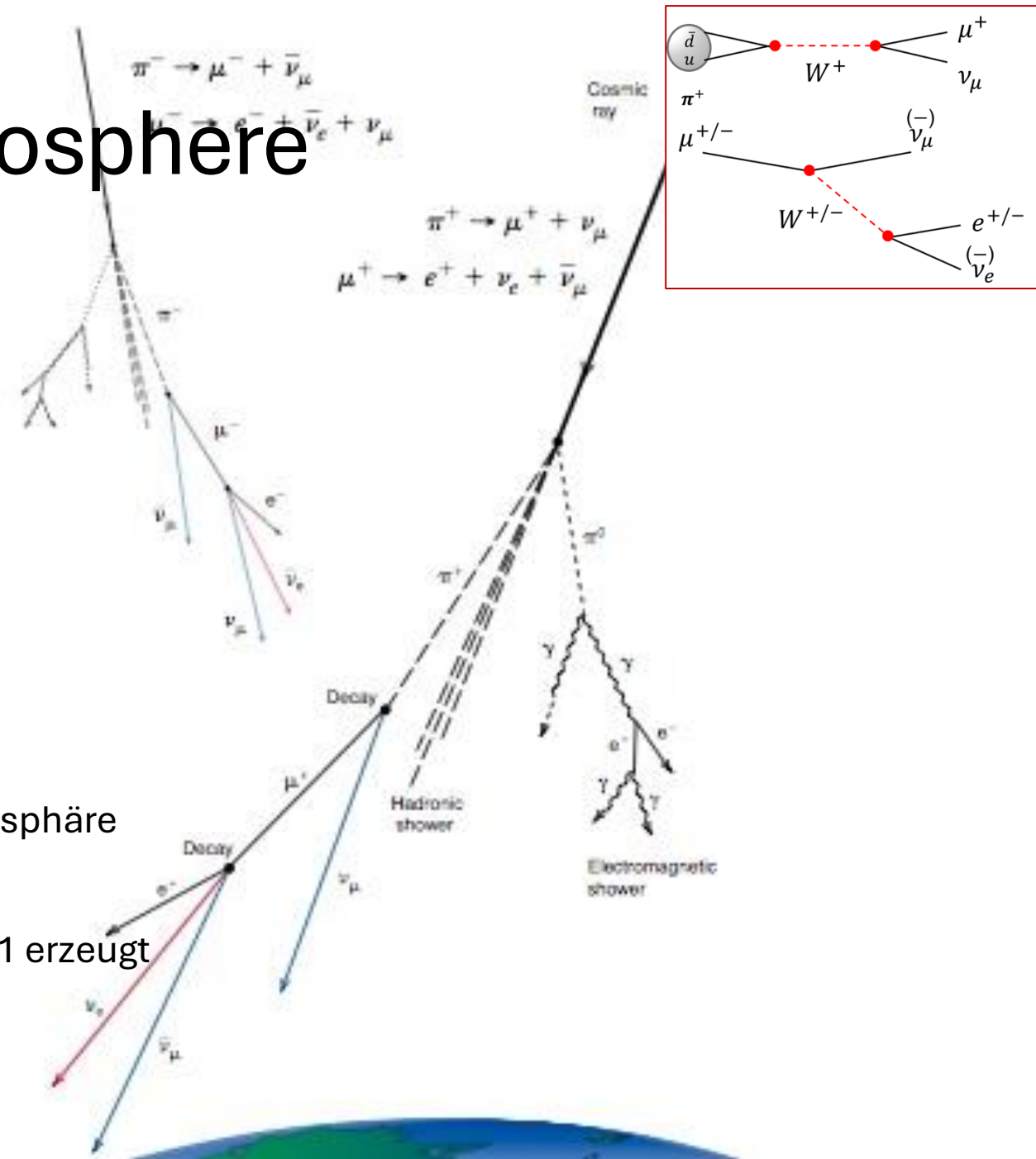


- Protonen treffen auf ein Target (z. B. aus Beryllium)
- Erzeugung von Pionen, Kaonen und charmhaltigen Mesonen
- Die Mesonen zerfallen und erzeugen Neutrinos

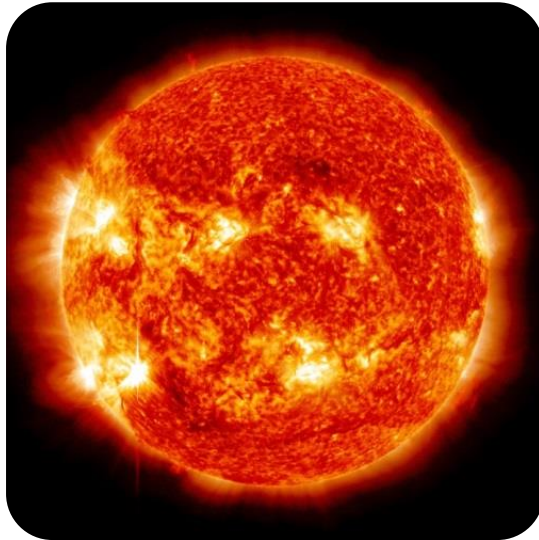
Neutrinos aus der Atmosphäre



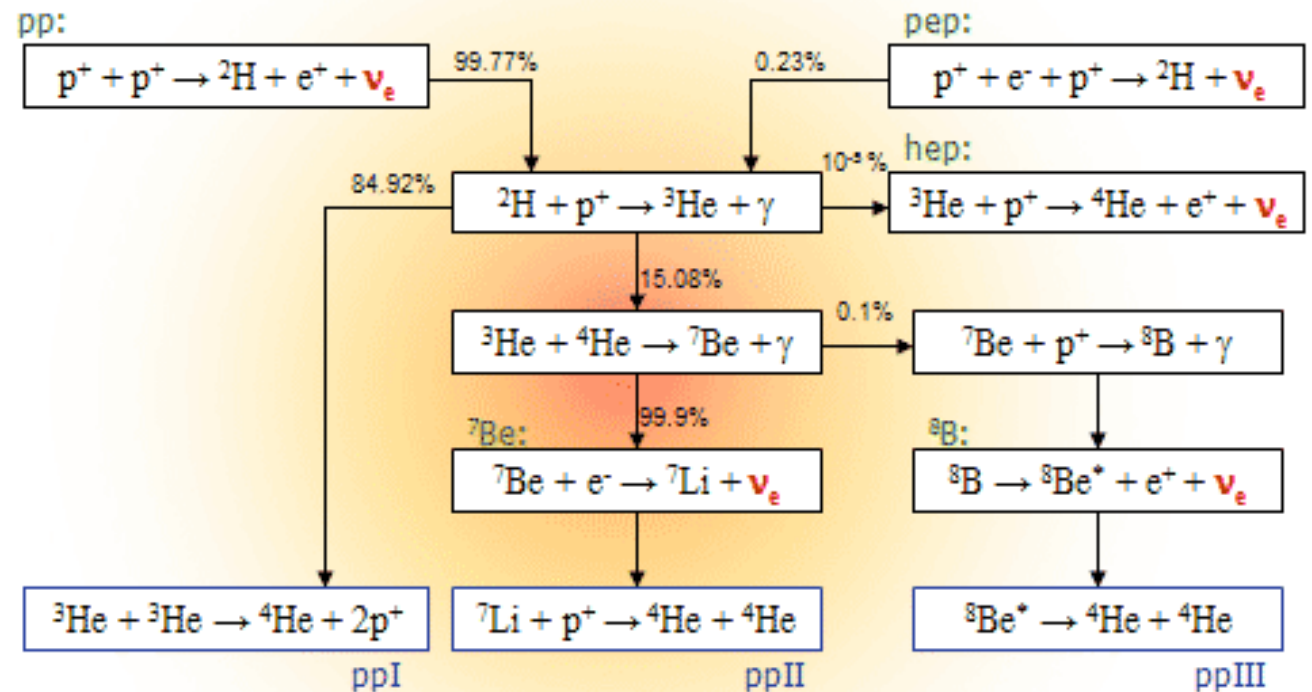
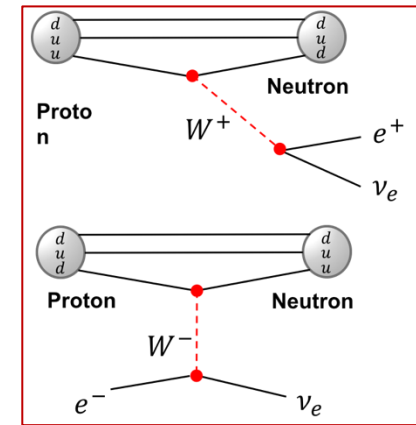
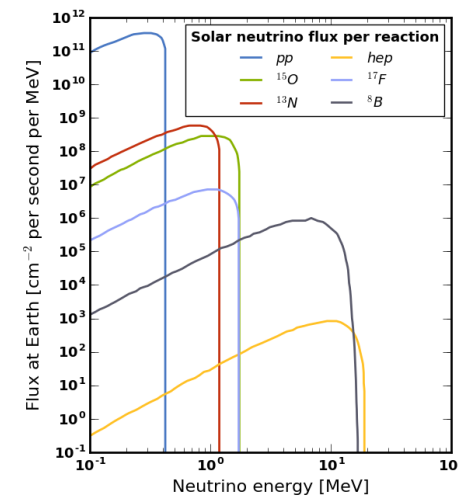
- Kollision kosmischer Strahlung (Protonen) mit der Atmosphäre
- Erzeugung von Schauer(n) aus Pionen (= ud-Quarks)
- Myon- und Elektron-Neutrinos werden im Verhältnis 2 : 1 erzeugt



Neutrinos von der Sonne

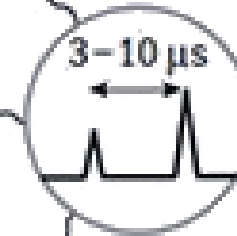
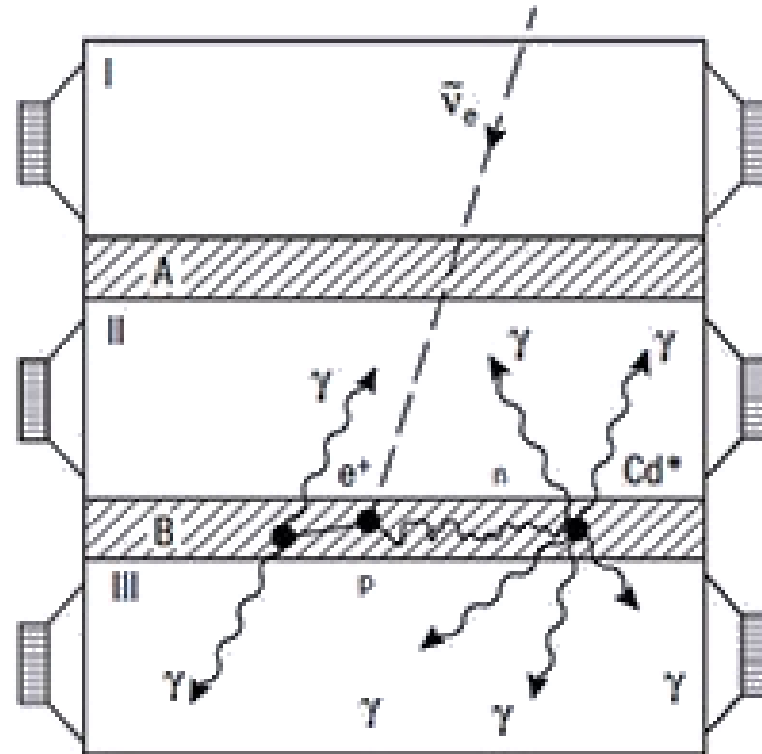
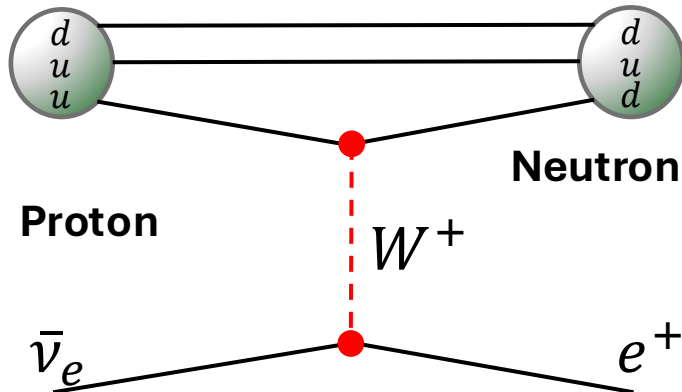
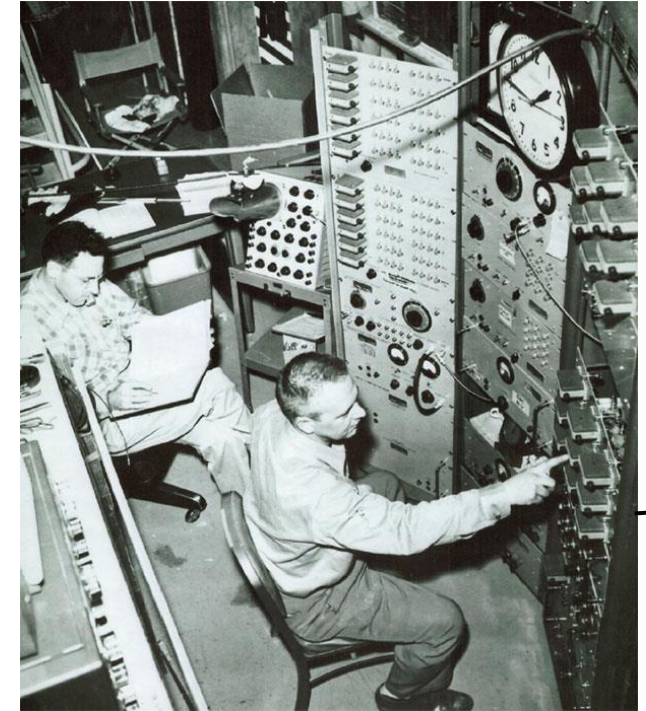


- Kernfusion in der Sonne
- Es werden ausschließlich Elektron-Neutrinos erzeugt
- 100 Milliarden (10^{11}) Neutrinos/cm²/s auf der Erde



Neutrino Entdeckung

- Neutrinofluss : $5 \times 10^{13} \bar{\nu}_e / \text{s/cm}^2$
- $\bar{\nu}_e$ wechselwirkt mit Proton in Wasser, und erzeugt ein Positron und ein Neutron

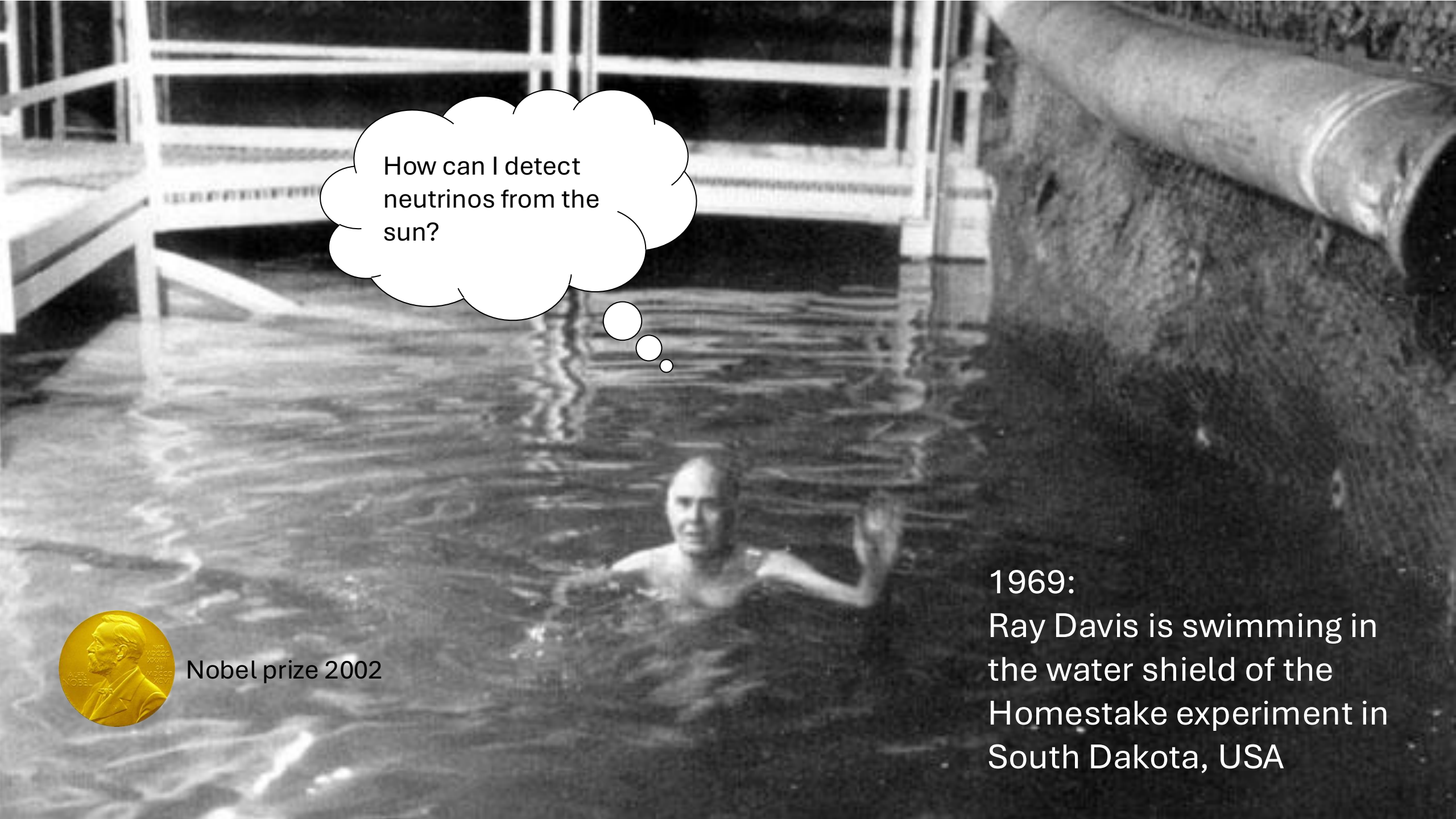


- Detector placed 11 m from reactor and 12 m underground
- 200 liters of water with about 40 kg of dissolved CdCl_2

Neutrino oscillations



Why?



How can I detect
neutrinos from the
sun?

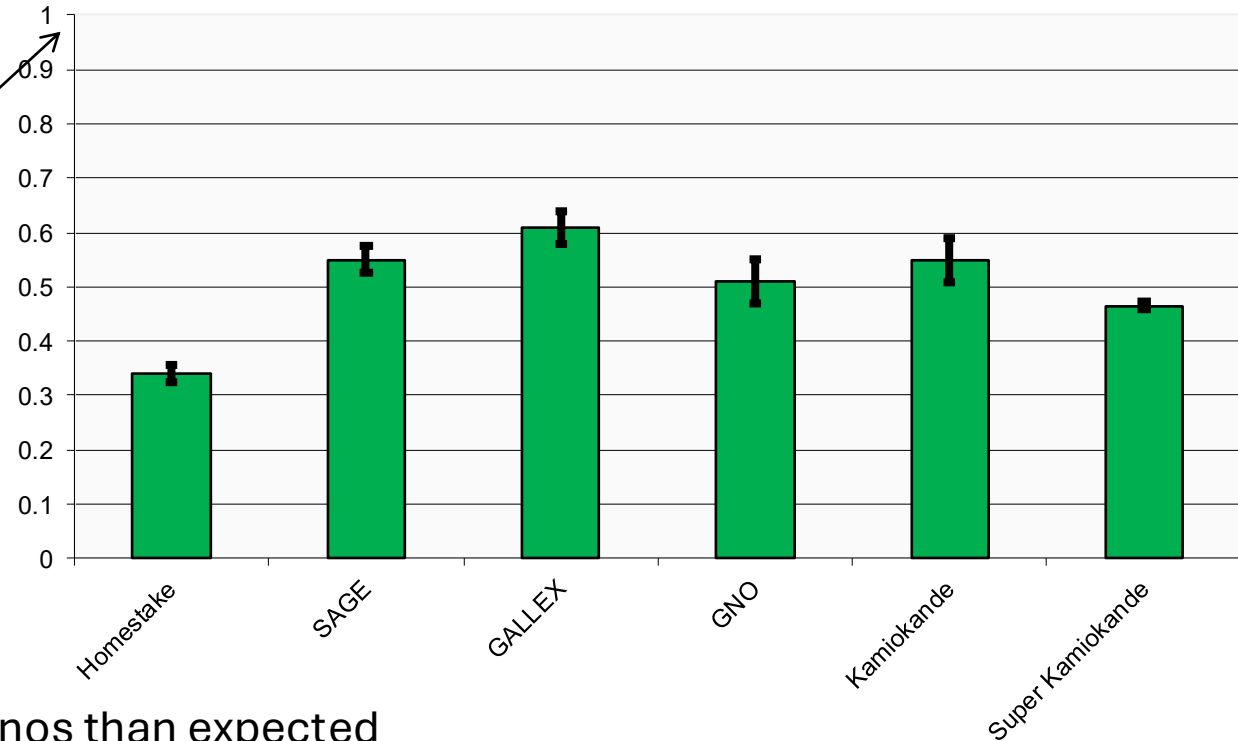


Nobel prize 2002

1969:
Ray Davis is swimming in
the water shield of the
Homestake experiment in
South Dakota, USA

Solar Neutrino Problem

Prediction of
John Bahcall



- ...All experiments measure less neutrinos than expected
- What is wrong? The expectation? The measurement?
- Or did the neutrinos change their flavor on the way from the sun to the earth?

Neutrino oscillations



What !?

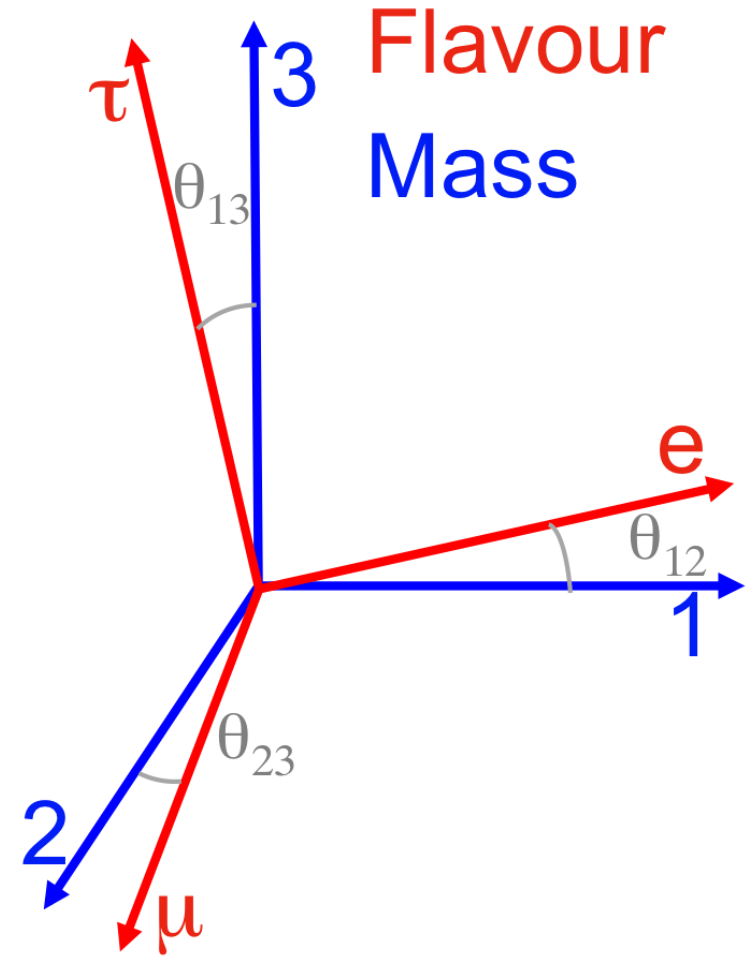
Neutrino Flavours und Masses

Flavor eigenstate =

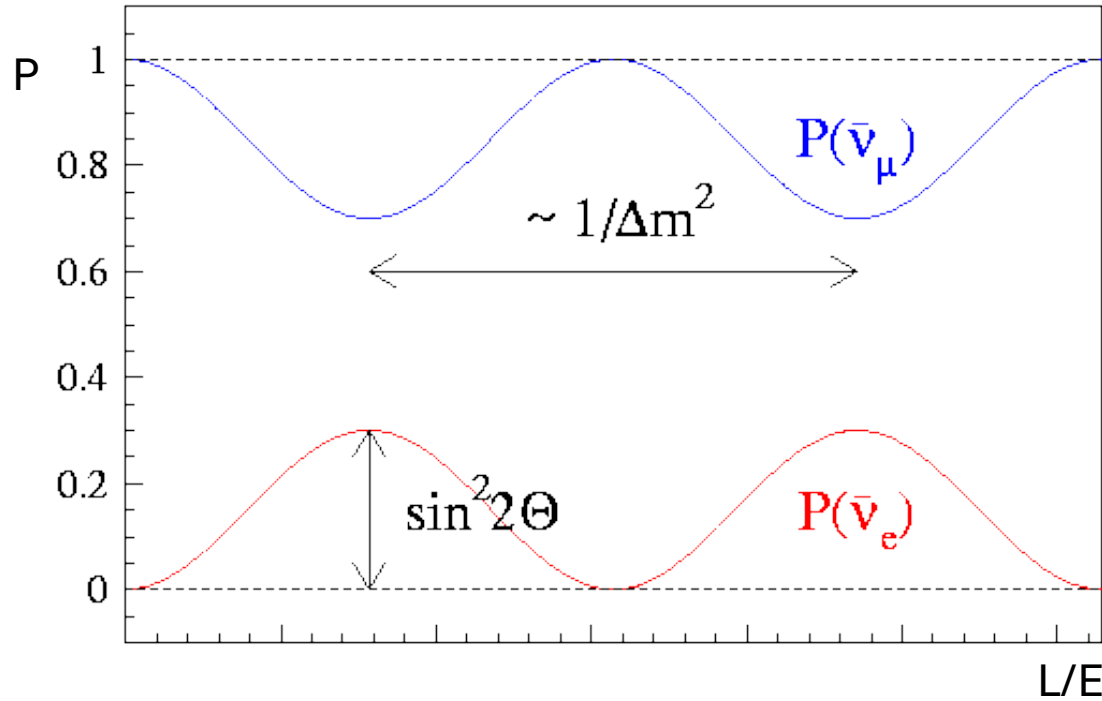
quantum mechanical superposition of **mass eigenstates**

Pontecorvo–Maki–Nakagawa–Sakata
Matrix describes the rotation:

$$\begin{pmatrix} \nu_e \\ \nu_\mu \\ \nu_\tau \end{pmatrix} = \begin{pmatrix} U_{e1} & U_{e2} & U_{e3} \\ U_{\mu1} & U_{\mu2} & U_{\mu3} \\ U_{\tau1} & U_{\tau2} & U_{\tau3} \end{pmatrix} \begin{pmatrix} \nu_1 \\ \nu_2 \\ \nu_3 \end{pmatrix}$$

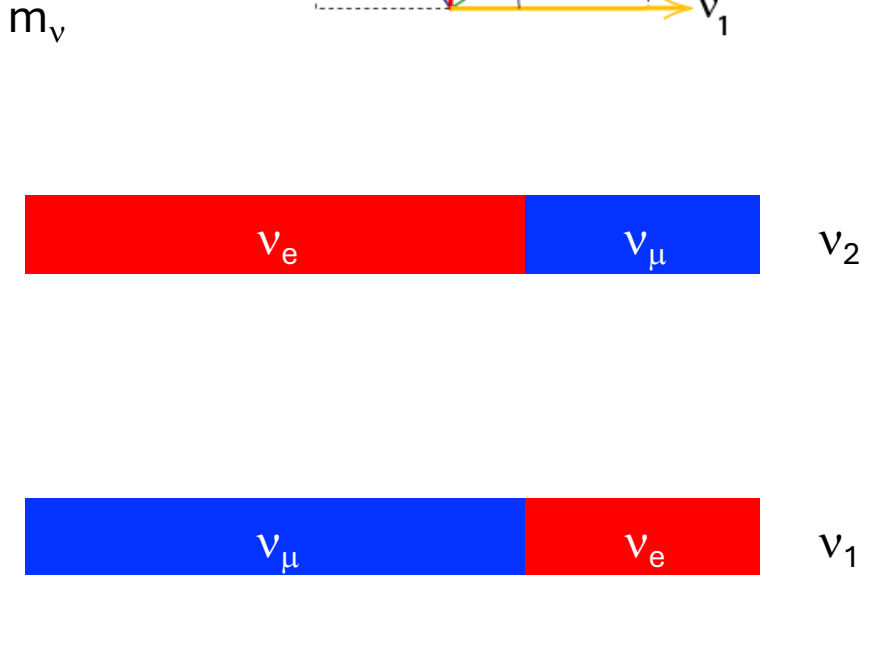
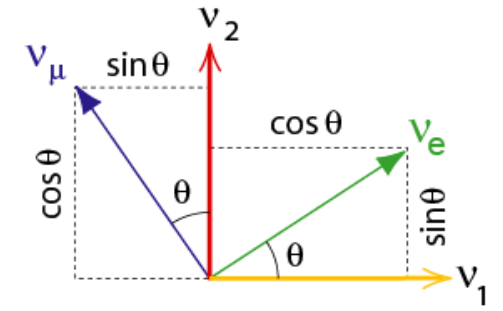


Neutrino Oscillations (for 2 Flavour)

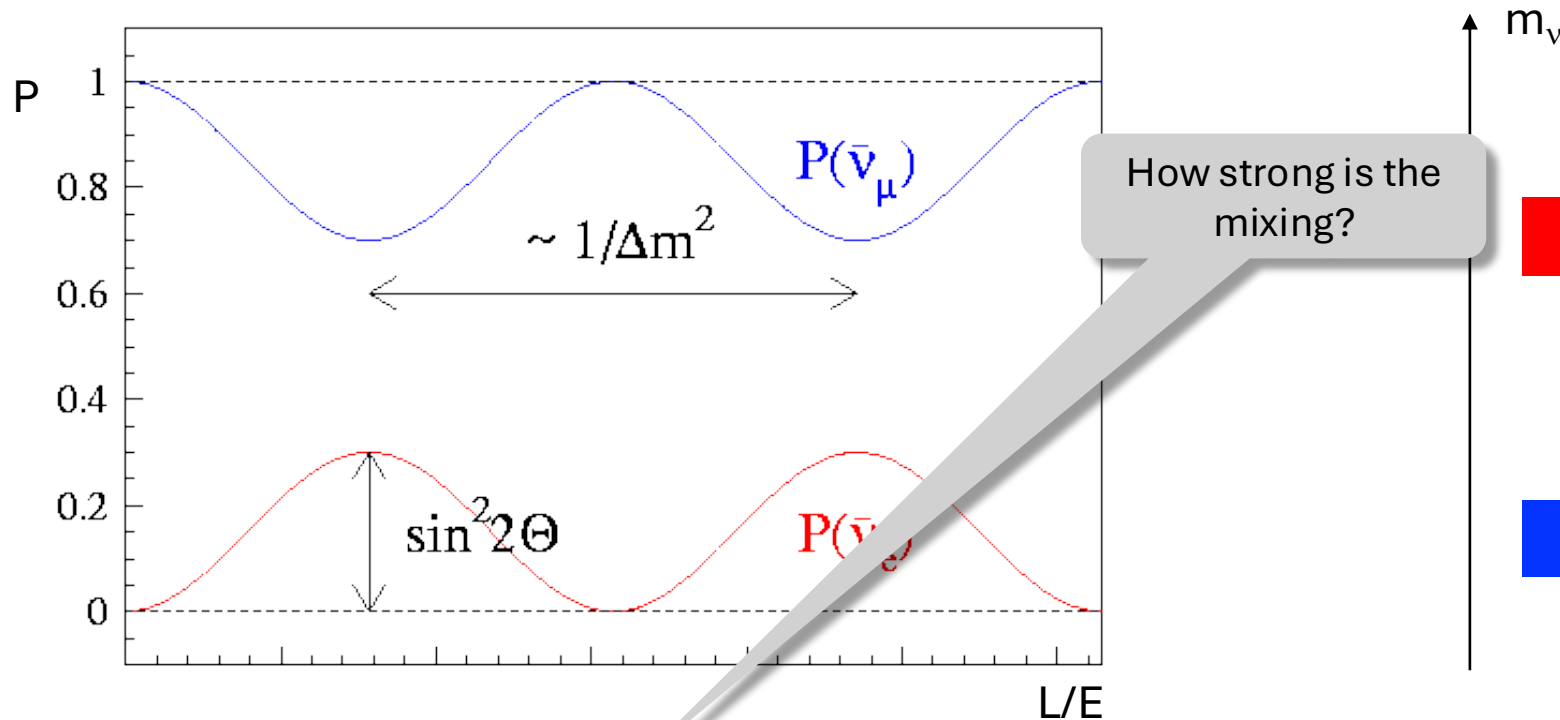
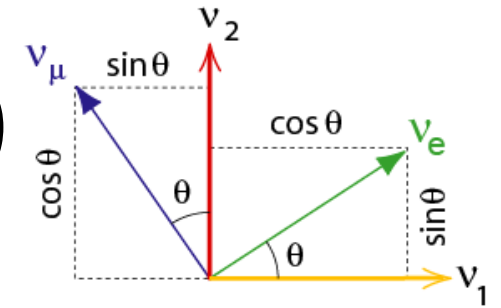


$$P(\nu_\mu \rightarrow \nu_e) = \sin^2 2\theta \cdot \sin^2(\Delta m^2 \cdot L_\nu / E_\nu)$$

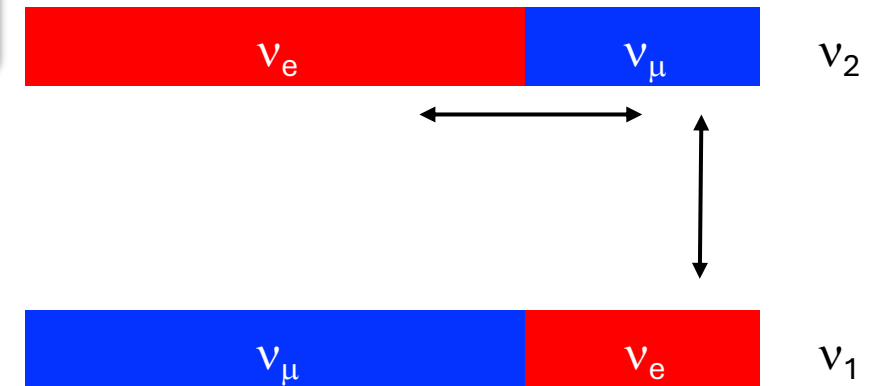
$$\Delta m^2 = m_1^2 - m_2^2$$



Neutrino Oscillations (for 2 Flavour)



How strong is the mixing?



What is the mass difference?

$$P(\nu_\mu \rightarrow \nu_e) = \sin^2 2\theta \sin^2(\Delta m^2 \cdot L_\nu / E_\nu)$$

Amplitude

Frequency

$$\Delta m^2 = m_1^2 - m_2^2$$

Neutrino oscillations are only possible if at least one neutrino has a mass

Neutrino oscillations

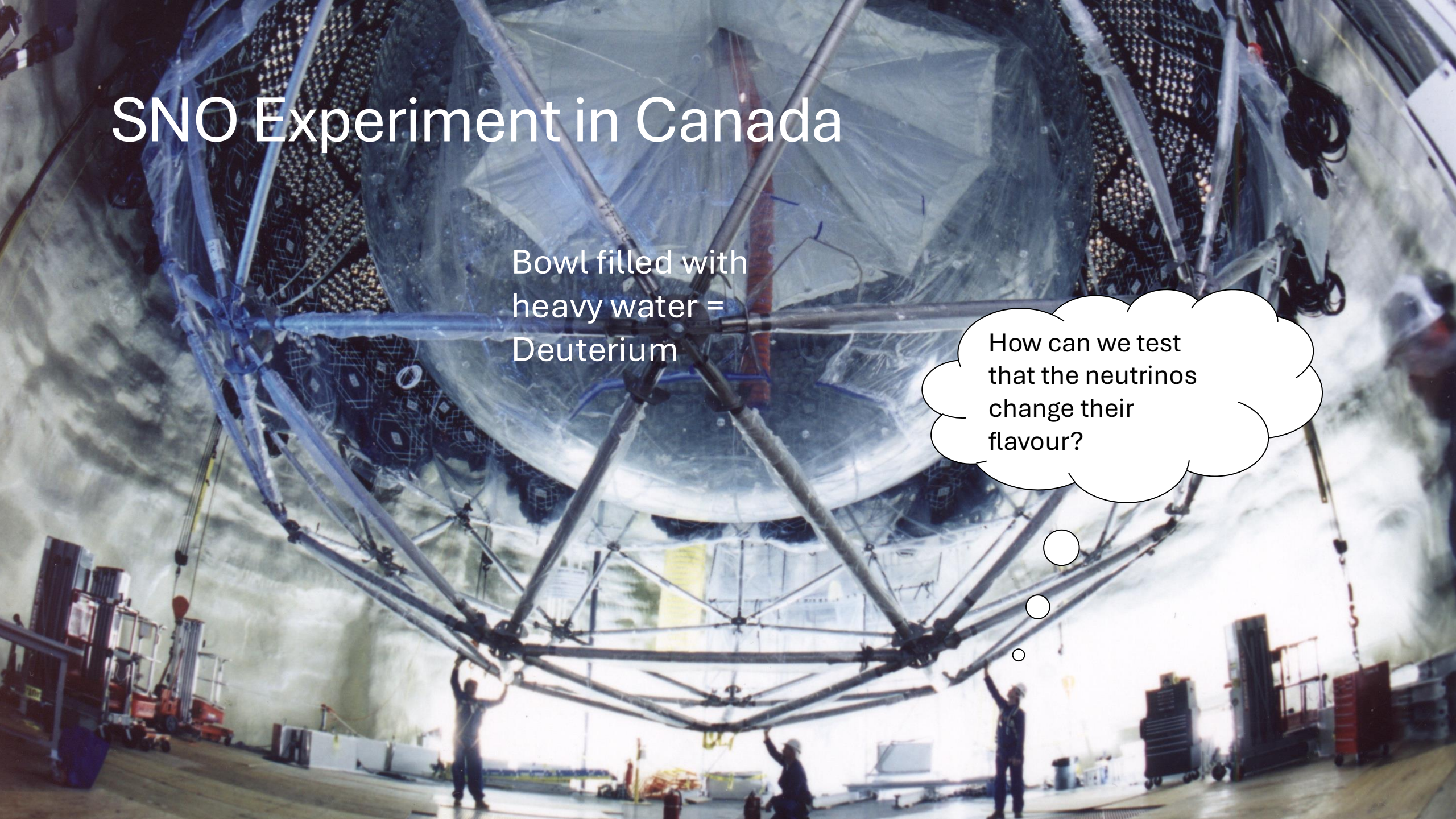


Really !?

SNO Experiment in Canada

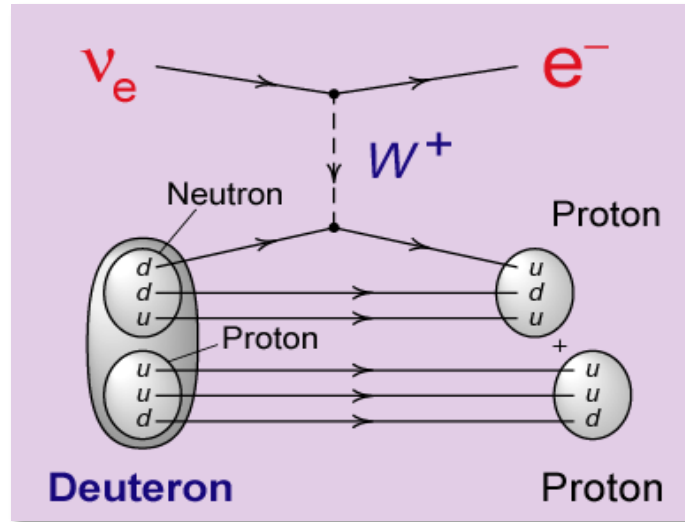
Bowl filled with
heavy water =
Deuterium

How can we test
that the neutrinos
change their
flavour?



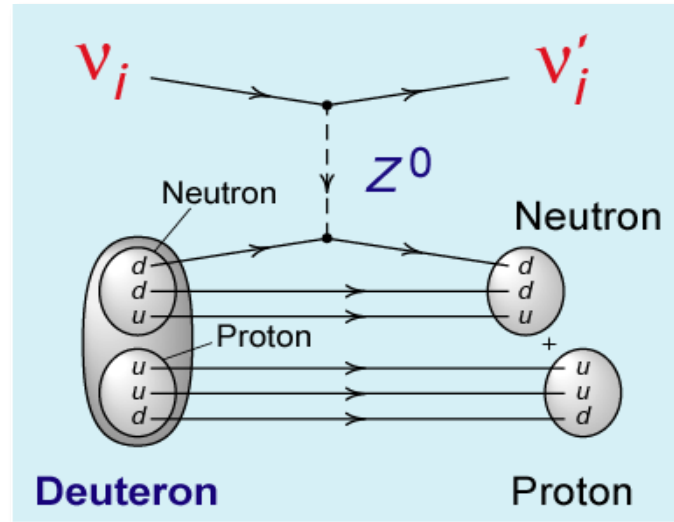
Idea!

charged current (CC)



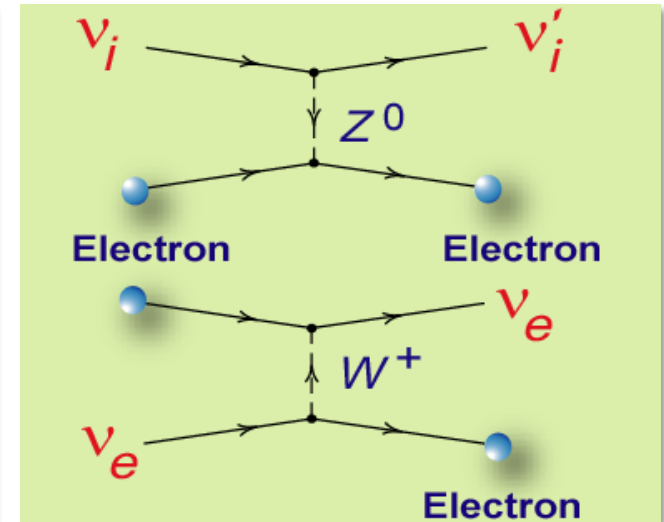
Only ν_e

neutral current (NC)



All Flavours

elastic scattering (ES)



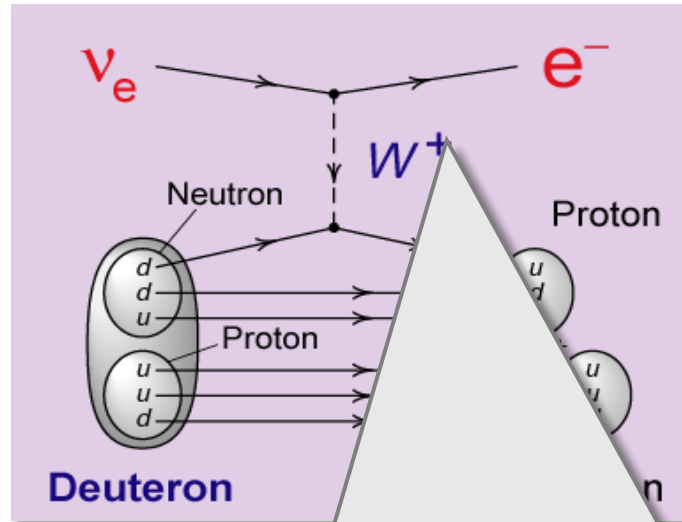
mostly ν_e

- Scattering via neutral Z-Boson is flavor independent
- This reaction channel measures the entire neutrino flux

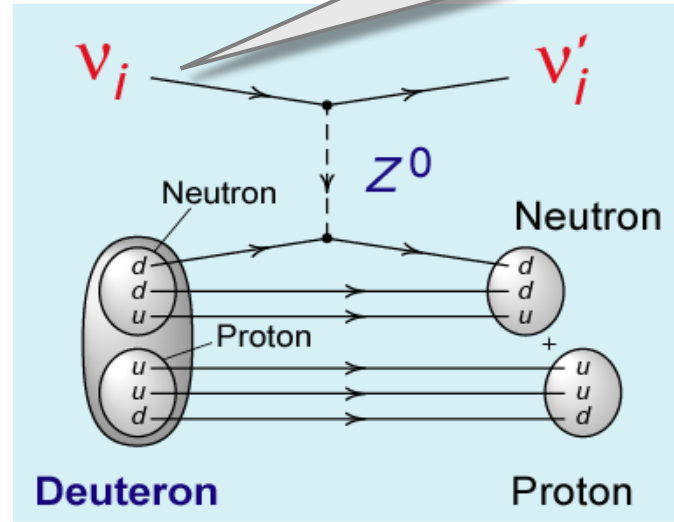
Idea!

- Binding energy of deuterium: 2.2 MeV
- Energy of neutrinos is high enough

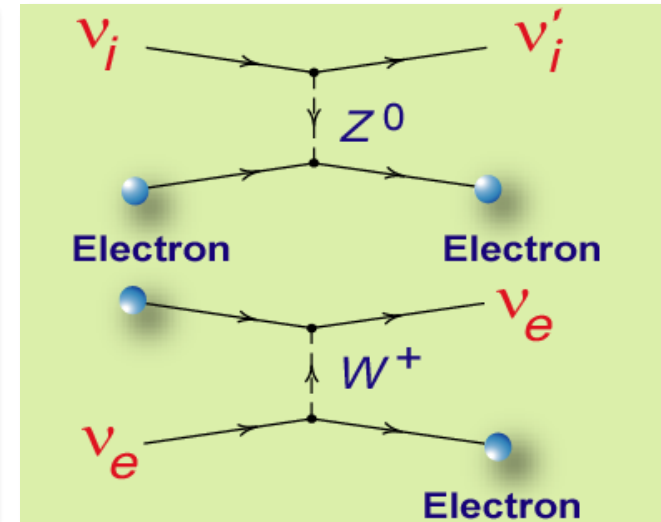
charged current (CC)



neutral current (NC)



elastic scattering (ES)

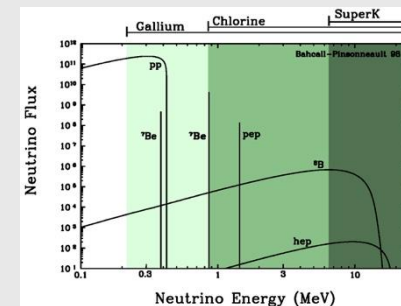


Only ν_e

All Flavors

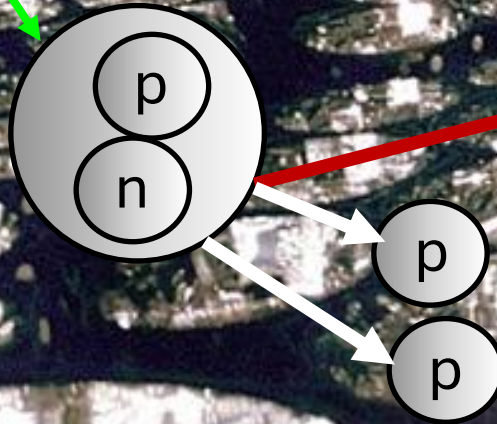
mostly ν_e

- For a CC reaction with a muon neutrino, a muon would have to be created
- $m_\mu = 105 \text{ MeV}$
- Electrons from the sun have smaller energies



SNO Phase 1: only heavy water

ν_e



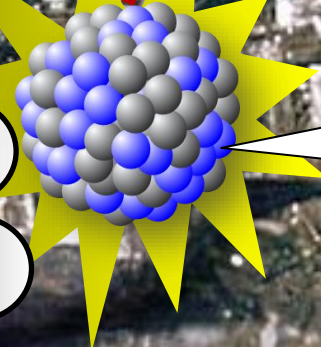
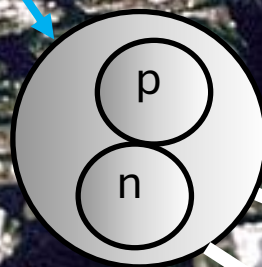
Cherenkov ring

Photomultiplier

SNO Phase 2: with 2t NaCl (Salt)

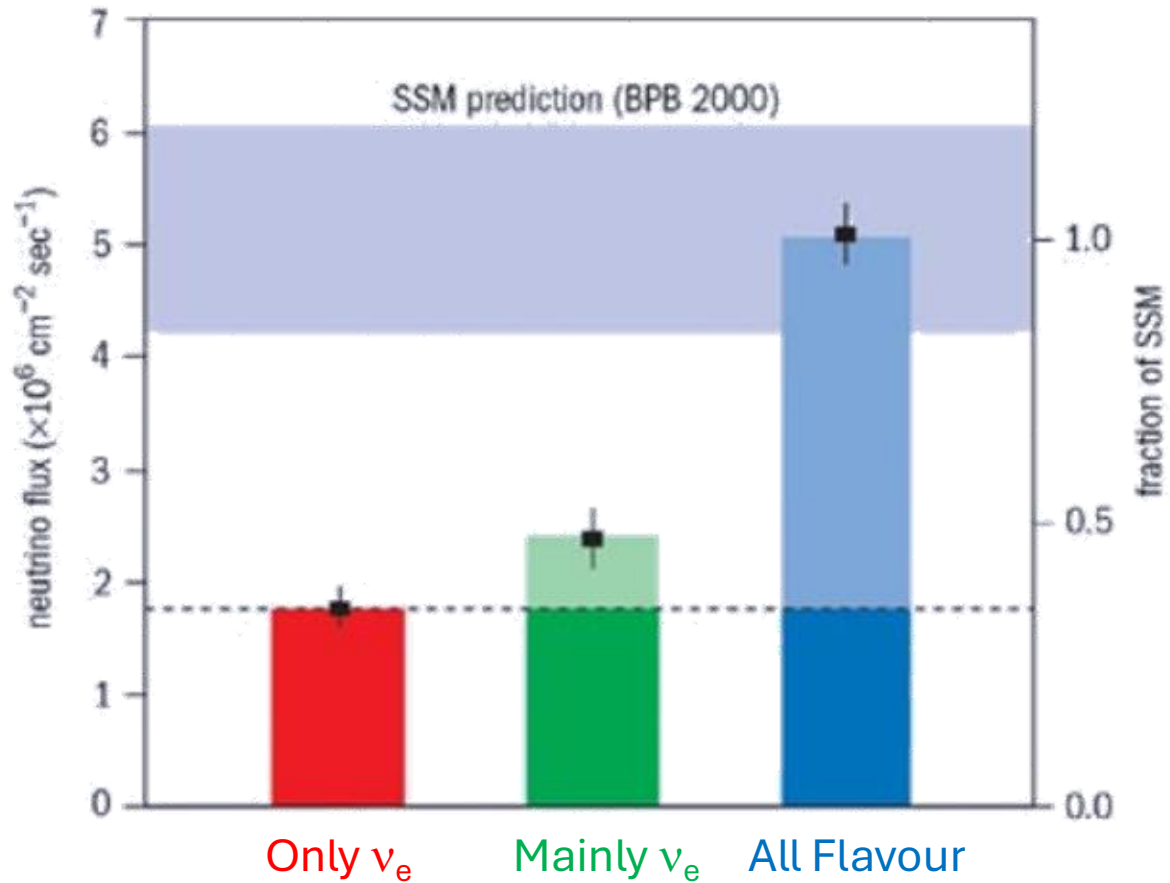


ν_e, ν_μ, ν_τ



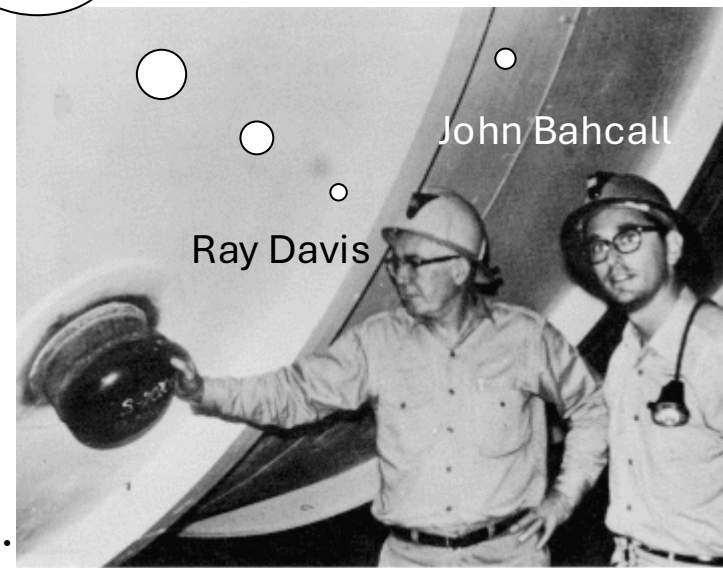
- Cl captures neutron
- $\text{Cl}^* \rightarrow \text{Cl}$ emits photons
- Photons kick electrons out of the atomic shell

...and what did SNO find?



So my
calculations
were right...

... and my
measurements,
too



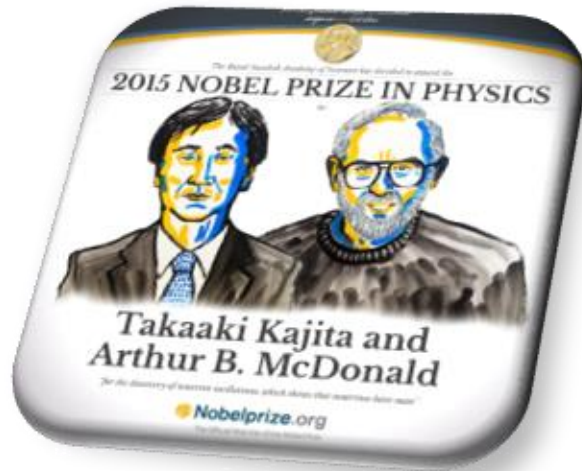
2001 (30 years later)...

Nobel Prize in Physics 2015



Takaaki Kajita and Arthur B. McDonald:

“for the discovery of neutrino oscillations, which shows that neutrinos have mass”



Prüfung

- **Inhalt**

- **Alle Vorlesungen** (Tafel und Slides)

- Schwerpunkt auf Verständnis!
 - Herleitungen werden nicht abgeprüft (z.B. Yukawa, Wirkungsquerschnitte)
 - Komplizierte Formeln (z.B. Wirkungsquerschnitte, Weizsäcker, Weißkopf Näherung) werden angegeben.
 - Abhängigkeiten und fundamentale Zusammenhänge sind prüfungsrelevant (z.B. Propagator Term, Energie-Impuls-Beziehung, de-Broglie Wellenlänge, Unschärferelation, Helizität...)
 - Wichtig: Auswahlregeln, Erhaltungssätze, welche Reaktionen sind erlaubt/verboten, Feynman Diagramme

- **Alle Übungen**

- Blatt 2 2a - Herleitung des Yukawa Potentials muss nicht im Detail gerechnet werden können.
 - Komplizierte Formeln werden angegeben.
 - Einheiten, Massen etc. werden angegeben.

- **Hilfsmittel**

- Taschenrechner (nicht programmierbar)
 - 1 DIN A4 Blatt (handschriftlich und selbst beschrieben, auch auf Tablet, kein TeX oder Computerschrift, kein Copy&Paste von Plots, Bildern etc.)